KEYWAVE

KW007-SE

5.8GHz Motion Sensor Module Datasheet

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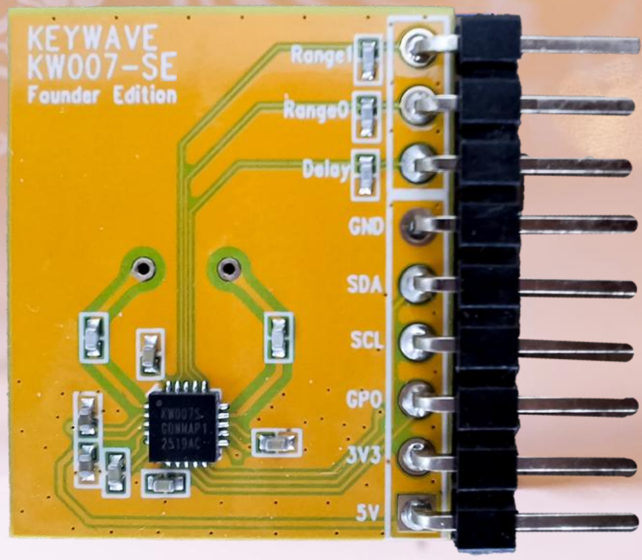
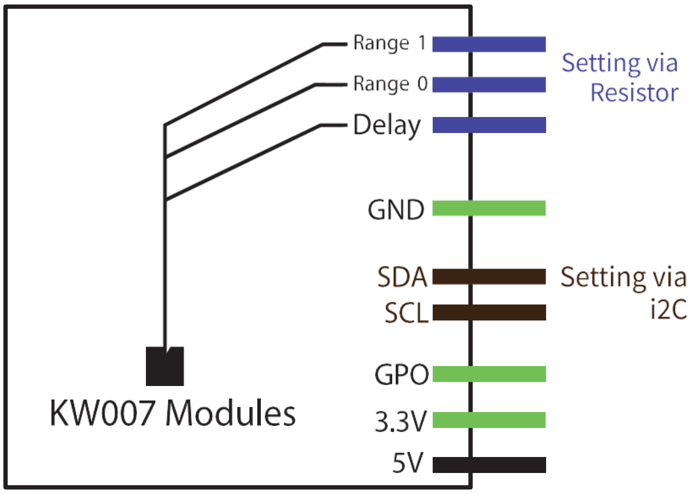
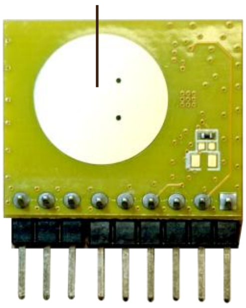
Keywave Technology Ltd.

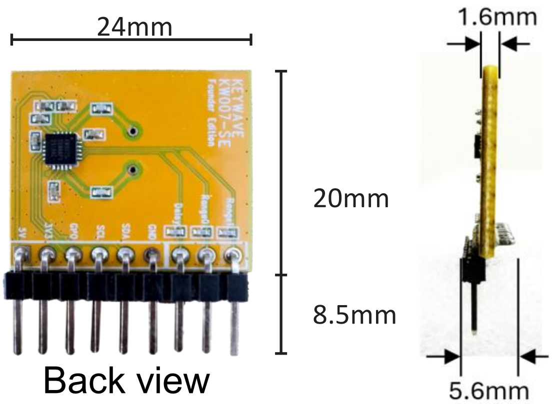




Sensor Antenna



Front view

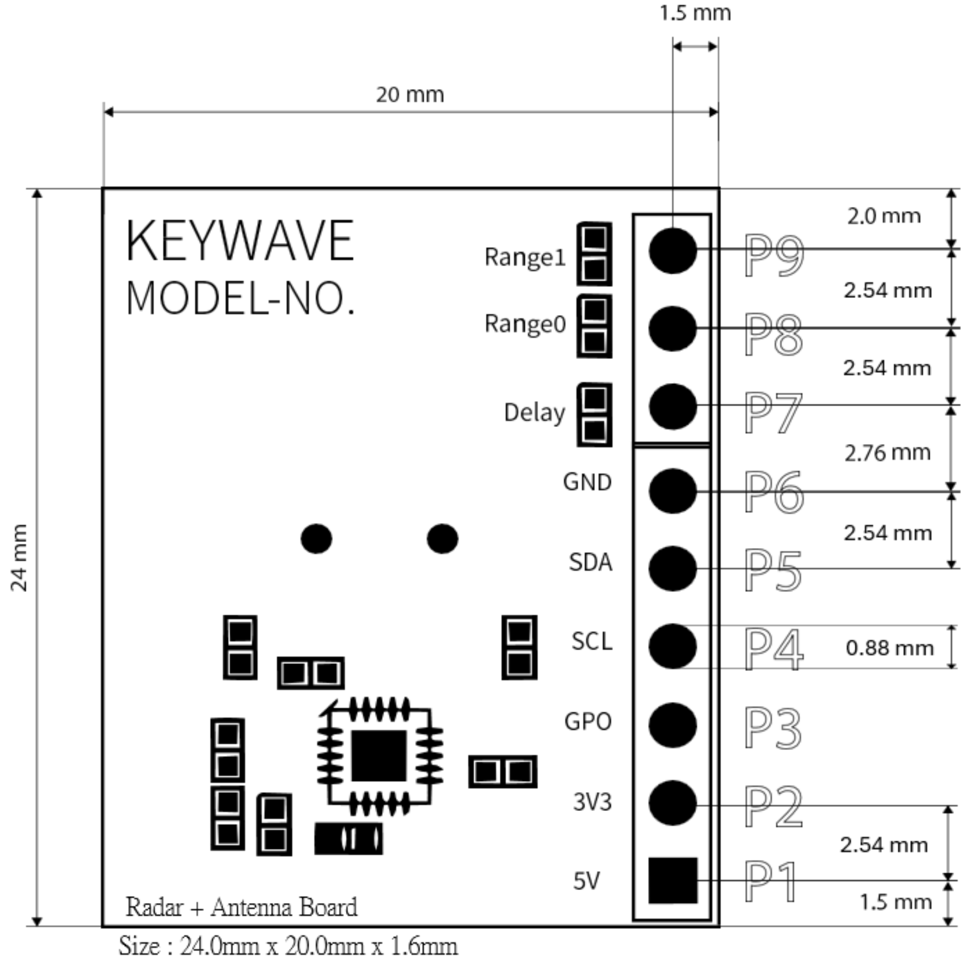


Back view

Side view

- **KW007-SE** is a high-performance, crystal-less 5.8GHz plug-and-play Doppler radar module (1T1R). Engineered for maximum reliability.
- World class ultra-low power consumption (30uA-100uA) in a compact form factor.
- **Scalable Power & Range:** Detection is adjustable from **10 cm to 12 m+**, with power consumption scaling from **30 µA** (short-range) to **100 µA** (long-range).
- **Dual Configuration Modes:**
  - ➢ **Standalone:** Set range and **hold-time duration** via resistors (no coding required)
  - ➢ **I2C Interface:** Unlock granular software tuning and dedicated **Approach/Leaving flags**.
- **Wide Coverage:** Broad **120° field of view** with a high-gain integrated antenna.
- Engineered for maximum reliability with **superior immunity to fan clutter, wind, and vibrations.**
- Recommended clearance zone: at least 8mm in total (front and back).

Module Dimensions & Pin Definitions



Description	Min.	Typ.	Max.	Unit
Module Length	23.98	24	24.02	mm
Module Width	19.98	20	20.02	mm
Module Thickness	1.57	1.6	1.63	mm
Hole Dimension	0.875	0.88	0.9	mm
Pin Header Diameter		0.6		mm

Pin No.	Pin Name	Description
P1	5V	5 V supply input
P2	3V3	3.3 V supply input (default) or LDO output
P3	GPO	General-purpose output
P4	SCL	I²C clock
P5	SDA	I²C data
P6	GND	Ground
P7	Delay	GPO extension time setting pin
P8	Range0	Radar detection range setting pin
P9	Range1	Radar detection range setting pin

- Power: 3.3 V or 5V: **Connect only ONE power supply (3.3V or 5V), do not connect both.
- 5V Input Mode: Input voltage is regulated by an on-chip LDO to power the internal core.
- 3.3V Input Mode (default): Bypasses the internal LDO. A low-noise power source is mandatory (an external LDO is recommended).
- It is recommended to add a **10μF decoupling capacitor** to power input.

Block Diagram

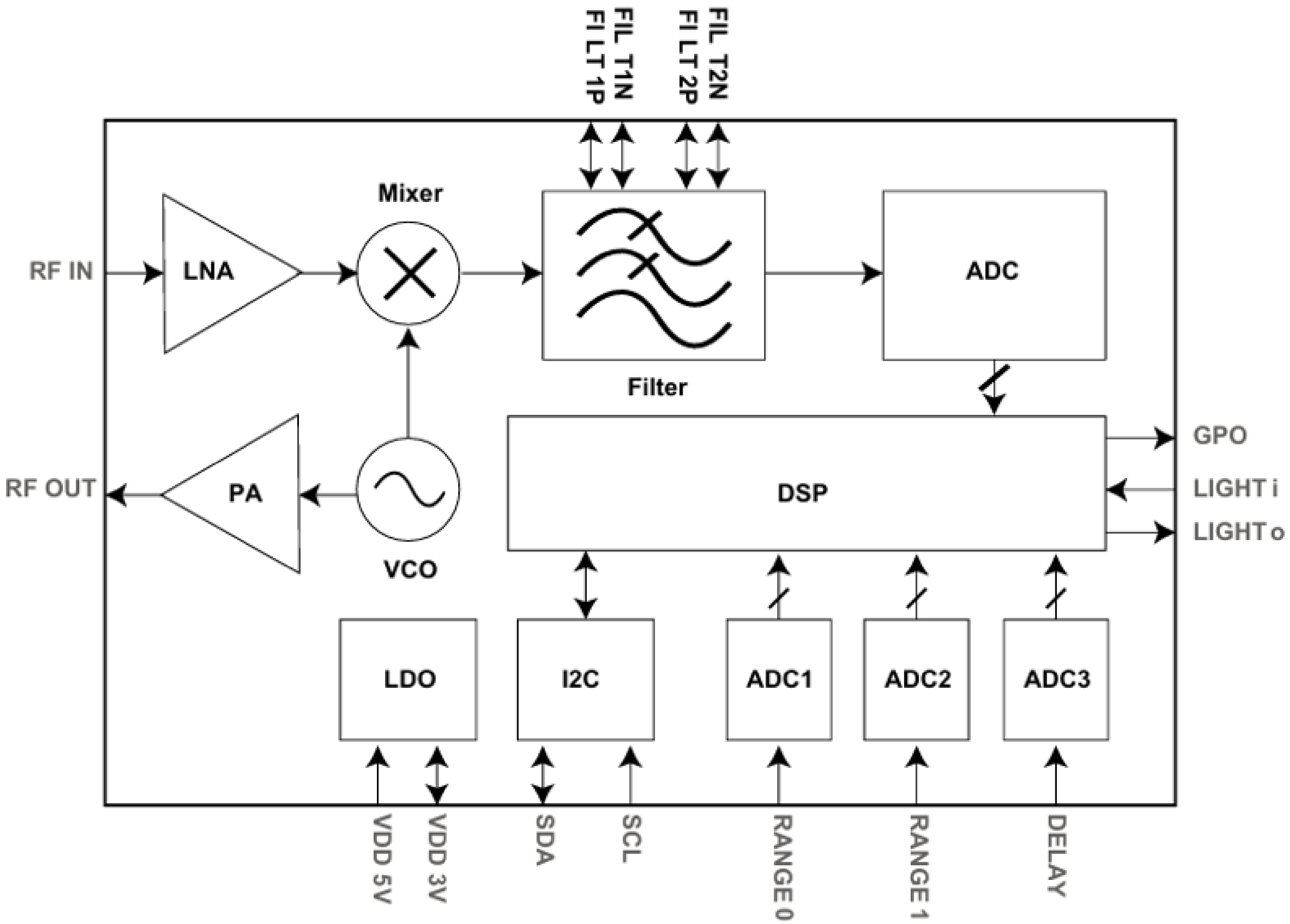


Figure : KW007 Block Diagram

Electrical Specification

Absolute Maximum Ratings

Characteristic	Symbol	Min.	Typ.	Max.	Units
5V supply voltage	V_{DD5V}	-0.3	-	5.5	V
3V supply voltage	V_{DD3V}	-0.3	-	3.6	V
Digital I/O voltage	V_{DIO}	-0.3	-	3.6	V
Digital I/O maximum current capability	$I_{DIO,max}$	-	-	4	mA*
Ambient Temperature	T_A	-25	-	85	°C

* It is recommended to add a 1 kΩ series resistor to the GPO pin to minimize current/voltage spikes.

* GPO cannot directly drive a high-current load; it requires a power MOS switch to do so.

DC Characteristics

Characteristic	Symbol	Min.	Typ.	Max.	Units
Supply voltage via 5V pin	V_{DD5V}	3.3	5.0	5.5	V
Supply voltage via 3.3V pin	V_{DD3V}	2.5	3.3	3.6	V
Operation Current	I_{DC}		30	100	uA

* It is recommended to add a 10μF decoupling capacitor to power input.

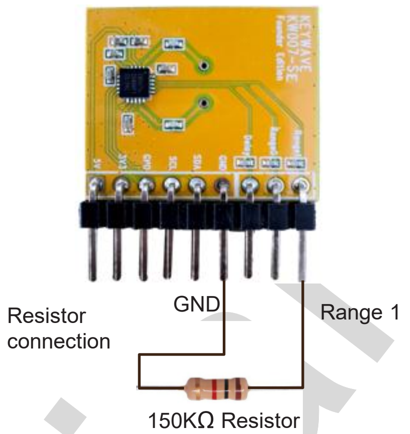
RF Characteristics

Characteristic	Symbol	Min.	Typ.	Max.	Units
Operation frequency	F_{RF}	5.725	-	5.875	GHz
RF output power	P_{RF}	-	2	-	dBm
Detection range	Range	-	-	12*	m

* Up to 12 m (requires I2C interface configuration).

* Horizontal alignment at 1m height.

KW007 Configuration via Resistors



- ## Range setting

Set Distance	Level 1	Level 2	Level 3	Level 4	Level 5	Level 6	Level 7	Level 8	*All Max
Range 0	0	150KΩ	270KΩ	*Open	0	150KΩ	270KΩ	*Open	*Open
Range 1	0	0	0	0	150KΩ	150KΩ	150KΩ	150KΩ	*Open
KW007-SE	0.1m	0.2m	0.3m	0.5m	0.7m	1m	2m	4m	4m

* 0 = connect to GND
* Open = no resistor fitted (∞ Ω) and not connect to GND.

***Note on Detection Range:** Listed distances are maximum reference values. Since detection is based on reflection signal strength, the actual range may vary depending on environmental factors, object size, and the material or thickness of the product housing.

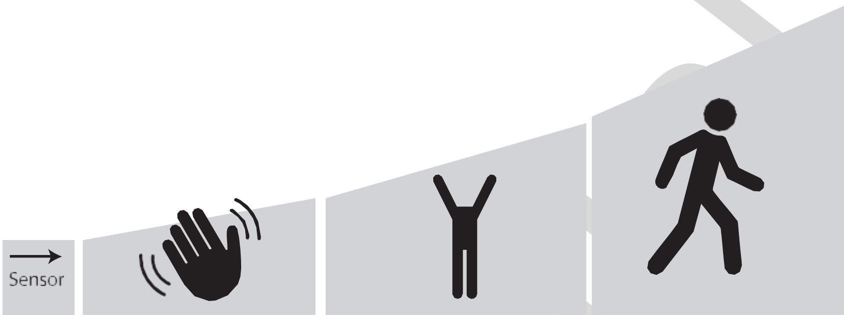
- ## Hold time setting

Set Delay	Max	Setting 1	Setting 2	Setting 3
Connect Resistor:	Resistor Open	Resistor 270KΩ	Resistor 150KΩ	Resistor 0
KW007-SE	64 sec.	1 sec.	32 sec.	16 sec.

***Hold-Time Logic:** When motion is detected, the GPO output pulls high. The output remains high for the duration of the set hold time. If motion is detected again while the output is already high, the hold timer will be re-triggered to extend the high-level duration.

• Detection Characteristics

Our sensor is designed to detect active motion. At close range, the sensor is highly sensitive and can detect small gestures like a hand wave. However, as the detection range increases, the environment becomes more complex. To maintain stability and prevent false triggers, targets at a greater distance require a larger displacement to trigger the sensor (e.g., walking or moving 50cm or more).



Longer ranges require larger movements for stable detection.

• Custom Configuration Via I²C

Once connected via I2C, the module unlocks the following advanced capabilities:

- **Directional Detection:** Access internal real-time flags to distinguish between **Approaching** and **Leaving** motion.
- **Detection Range:** It can reach up to 12 meters.
- **Precision Tuning:** Digitally calibrate detection sensitivity with granular accuracy.
- **SDK & Support:** Visit **www.keywavetech.co.uk** to download the official SDK and integration guides.

• Ordering & Custom Builds

100k+ units: Specify your desired detection range & pulse duration, and we'll pre-program your modules.

IC-Only Purchases: available on request—please contact your Keywave sales representative.
or email to : W@keywavetech.co.uk

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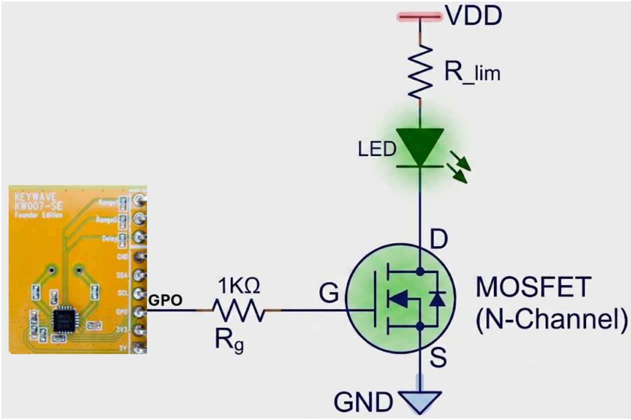
2. A critical component is any component of a life support device or system whose failure to perform can be reasonably expected to cause the failure of the life support device or system, or to affect its safety or effectiveness.

Application Guide



* LED driver circuit with pin GPO.

GPO Application Note :

- GPO cannot drive an LED directly; it requires a power MOS switch with a 1 KΩ series gate resistor to do so.

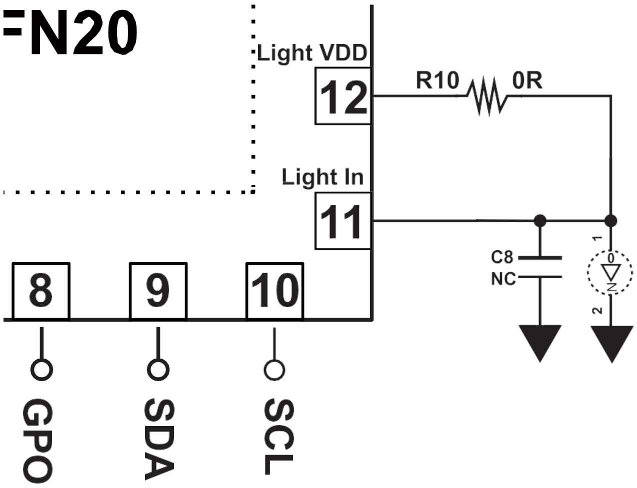
- Power off before plugging or unplugging the KW007. Do not hot-plug.

Power Supply Note:

- Decoupling: If using long leads or bench supplies, place low-ESR capacitors (e.g., 10μFelectrolytic/tantalum in parallel with 0.1μFceramic) as close as possible to the module's power input pins to suppress lead inductance and stabilize voltage.

- When testing with a USB power bank, please ensure using a cable rated for high current delivery.

Appendix :Ambient light sensing



Short Pin12 to Pin11 (R10=0R) to turn off light sensing function

This circuitry is used to implement the ambient light sensing function on our module. Recommended component values are as follows:

R10 : 1 MΩ
C8 : 100 nF
U4 : SLPT0805AC (ambient light sensor)